

# Control and Optimization for Modern Energy Systems

*A battery plant tour and an afternoon of technical talks at the University of Hawai'i at Mānoa*

| DATE                                                                      | TIME                                         | VENUE                                     | REGISTRATION                             |
|---------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------|------------------------------------------|
| <b>Monday, December 14, 2026</b><br>Adjacent to IEEE CDC 2026, Dec. 15–18 | <b>8:00 am – 6:00 pm</b><br>Tour, then talks | <b>UH Mānoa campus</b><br>Honolulu, O'ahu | <b>No fee</b><br>Separate event from CDC |

## About the workshop

The IEEE CSS Technical Committee on Energy Systems (TC-ES) is organizing a full-day technical workshop that brings together researchers and industry practitioners who work on control and optimization for modern energy systems.

A university campus in Hawai'i is a fitting setting for this workshop, because the islands operate some of the most renewable-rich, inverter-dominated grids in the world. Many of the problems that engineers in Hawai'i are solving today inform global solutions for grid-forming control, distributed flexibility, large new loads, and reliability under deep uncertainty.

## Program

- Grid-forming and inverter-based resource control
- Next-generation grid codes and ancillary services
- Grid-aware demand response and DER coordination
- Congestion and voltage management
- Virtual power plants and market participation
- Data centers and AI compute as flexible grid assets
- Operational perspectives from Hawaiian Electric

## Confirmed speakers



**Ken Aramaki**  
Director, Integrated  
Grid Planning,  
Hawaiian Electric



**Mark Glick**  
Chief Energy Officer,  
State of Hawai'i



**Daisy Green**  
University of Hawai'i  
at Mānoa



**Bai Cui**  
Iowa State University



**Sairaj Dhople**  
University of Minnesota



**Vladimir Dvorkin**  
University of Michigan,  
Ann Arbor



**Verena Häberle**  
ETH Zürich



**Patricia  
Hidalgo-Gonzalez**  
UC San Diego



**Sorin Olaru**  
CentraleSupélec



**Siva Seetharaman**  
Purdue University



**Johanna Vorwerk**  
Technical University of  
Denmark (DTU)

## Schedule

### 8:00 AM – 1:00 PM

#### Tour of the Kapolei Energy Storage facility

Buses take participants from UH Mānoa to Kapolei on western O'ahu to visit a 185 MW / 565 MWh grid-forming battery plant.

### 1:00 PM – 6:00 PM

#### Technical talks at UH Mānoa

Invited talks from universities, Hawaiian Electric, and the State of Hawai'i.

## Logistics

- UH Mānoa is a 15-minute drive or a 50-minute walk from the CDC hotel, the Hilton Hawaiian Village.
- The workshop is separate from CDC and has no registration fee.

## WORKSHOP ORGANIZERS

Daisy Green, Johanna Mathieu, Mads Almassalkhi (Chair, IEEE CSS TC-ES), and Patrick Panciatici

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Program and updates  
[madsalma.github.io/  
hawaii2026.html](https://madsalma.github.io/hawaii2026.html)